

Do Oil Shocks Hit Airlines Equally?



Charles Alvarez, John Gram, & Filipe Cancado

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Professor Jin Man Lee

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Introduction

Oil price shocks have long been recognized as an important source of macroeconomic and financial market volatility. Geopolitical conflicts, supply disruptions, and uncertainty surrounding major oil-producing regions can generate sudden increases in energy prices that affect firms, consumers, and investors worldwide. Recent tensions involving Iran and the Strait of Hormuz have renewed concerns about the vulnerability of global energy markets, as a significant share of the world's oil exports pass through this critical maritime chokepoint. Disruptions to these flows have the potential to increase crude oil prices and create substantial uncertainty across industries that depend heavily on fuel inputs.

Commercial aviation is an industry where these tensions may be a particular point of concern. Fuel expenses typically account for between 20 and 30 percent of an airline's operating costs. Airlines may experience declining profitability, lower expected cash flows, and reduced stock valuations. Differences in geographic location, fuel sourcing practices, hedging strategies, and exposure to regional energy markets may cause the financial impact of oil shocks to vary considerably across carriers. European carriers are heavily influenced by Brent crude prices, Asian carriers rely on fuel markets connected to Middle Eastern imports and regional refining capacity, and U.S. carriers are more closely tied to West Texas Intermediate (WTI), which may be partially insulated by domestic production.

This paper examines the effects of Iranian-conflict-driven oil price shocks on major airlines in North America, Europe, and Asia. This paper employs a combination of Structural Vector Autoregression (SVAR), local projection methods, event-study techniques, and GARCH volatility models. These approaches allow for the identification of oil supply shocks, estimation

of airline-specific stock market responses, and analysis of volatility dynamics during periods of geopolitical tension. The paper seeks to improve our understanding of how energy market disruptions are transmitted through the global aviation industry and to identify the characteristics that make some airlines more resilient to oil-price volatility than others.

Literature Review

Kilian and Park (2009) ask whether oil price shocks affect U.S. stock returns, and if so, whether the answer depends on what caused the price of oil to move in the first place. The motivation for this paper was to provide a fresh perspective as the link between oil prices and U.S. stock returns. They realized that there was not consensus in previous research on the relationship between stock prices and oil. While some studies found that oil price increases were associated with market declines, others suggested they had no effect at all, or that relationships were unstable. Most existing work treated crude oil prices as exogenous to the U.S. economy. The authors argue the opposite is true, and oil prices respond to the same global economic forces that drive stock prices. Previous models presumed that the impact of higher oil prices would be the same regardless of why the price increased. Because the relative importance of different types of shocks changes over time, the authors argue that older regressions were biased toward finding no significant relationship or relationships that were unstable over time.

The authors utilized a comprehensive set of monthly data spanning from January 1973 to December 2006 to analyze the relationship between oil shocks and the U.S. stock market. The variables in their regression were global crude oil production, a measure of real global economic activity, the real price of oil, and aggregate U.S. real stock returns. To estimate the impact of oil price changes on the U.S. stock market, they utilized a structural vector autoregression model

(SVAR) designed to disentangle the underlying causes of oil price fluctuations. They then imposed a block-recursive structure on the relationship between reduced form disturbances and underlying structural shocks. This allowed them to isolate oil market shocks into three categories: oil supply shocks, aggregate demand shocks, and oil-specific shocks.

The main findings of this paper are that the impact of oil price change on U.S. stocks returns depends fundamentally on the underlying cause of the shock. The authors conclude that the oil market demand and supply shocks jointly account for 22% of the long-run variation in U.S. stock returns. Where oil supply shocks have a statistically modest effect on aggregate stock returns, while demand shocks driven by geopolitical uncertainty produce significant negative stock responses in the short term. Particularly, they discovered considerable heterogeneity between industries impacted by oil shocks. With transportation and energy sectors responding differently from the aggregate market. This paper provides a direct methodological blueprint for our SVAR identification strategy. We will extend this framework to airline specific stock returns across global markets, allowing us to test the demand shocks most relevant to an Iranian conflict scenario.

Jordà (2005), introduces a method for calculating the dynamic effects of economic shocks. Traditionally, researchers use vector autoregressions (VARs) to estimate impulse responses, but these models are often prone to misspecification and struggle with complex nonlinearities. Jordà proposes local projections, which involve running a series of individual regressions for each future time horizon instead of extrapolating from a single global model. This approach is more robust, and simpler to implement using regression software like R.

The central methodology is the use of local projections, which compute impulse responses without the need to specify and estimate a global multivariate dynamic system like VAR. The

core approach is that instead of extrapolating into distant horizons from a single model, local projections are estimated at each period of interest. This is done by running a sequence of regressions where the endogenous variable is shifted forward in time. The estimation equation for a local projection represents a sequential regression where variables of interest are projected directly onto their own past values for a specific future timeframe. This is the defining feature in the methodology that it does not use a single model to predict the entire future; instead, we can estimate a collection of separate regressions, one for every single period of interest.

The main finding of the paper is that it establishes local projections as a robust, flexible, and statistically simple alternative to VAR for estimating impulse responses. Furthermore, using Monte Carlo simulations found that the model using local projections was much more accurate than VARs at capturing the true patterns of impulse responses over time. The method also makes it straightforward to construct valid confidence bands at each individual horizon without the compounding uncertainty that accumulates when using VAR.

We implement local projections as a robustness complement to our SVAR estimates. The local projections framework is especially valuable in our setting because the relationship between geopolitical oil shocks and airline stock returns may be nonlinear, particularly during periods of acute conflict escalation, and because our relatively short sample makes lag uncertainty a genuine concern.

Engle (1982) introduces the Autoregressive Conditional Heteroscedastic (ARCH) process, a new class of stochastic processes where the variance of the current error term is conditional on the past. In the ARCH model, the conditional variance, the forecast of today's uncertainty, changes over time as a function of past squared errors, even while the unconditional variance remains constant. Motivated by the observation that financial and macroeconomic series

exhibit periods of high and low volatility that cluster together over time. This is a pattern that standard ARIMA models, which assume constant variance, cannot be captured.

Bollerslev (1986) extends Engle's framework proposing a natural generalization of Engle's work to allow for a more flexible and parsimonious description of variance. The Generalized ARCH (GARCH) process extends the ARCH model by allowing the current conditional variance to depend not only on past squared errors but also on past conditional variance.

The two papers utilize distinct but related methodologies and datasets to model time-varying volatility in inflation rates for the U.K. and the U.S. Engle employed Maximum Likelihood Estimation (MLE) to estimate the parameters, utilizing a scoring iteration method. To determine the presence of ARCH effects, he developed a Lagrange Multiplier (LM) test based on the sample size multiplied by the squared multiple correlation coefficient of a regression of squared OLS residuals on their own lagged values.

Bollerslev generalized Engle's work by proposing the GARCH process. This methodology allows the current conditional variance to depend not only on past squared errors, as in ARCH, but also on past conditional variance. His approach derived the formal conditions for wide-sense stationarity and the autocorrelation structure for this new class of models. For estimation, he utilized MLE via the Berndt, Hall, Hall, and Hausman (BHHH) algorithm, which is convenient for handling the recursive terms in the GARCH model. He also adapted the LM test to check for GARCH effects. The GARCH (1,1) specification, with a single lag of each term, proves sufficient to capture volatility clustering in most financial applications and becomes the workhorse model of empirical finance.

The main findings of the two papers center on the development and empirical validation of models that account for time varying volatility in economic data. Engle successfully introduced the ARCH process, a new class of stochastic processes that are serially uncorrelated with a mean of zero but possess nonconstant variances conditional on the past while maintaining a constant unconditional variance. Applying the model to U.K. inflation, Engle found the ARCH effect to be highly significant. While OLS remains consistent, MLE using ARCH is significantly more efficient and provides more realistic forecast intervals by better identifying clusters of large errors.

Bollerslev found that GARCH models provides a more parsimonious description of the variance, using fewer parameters, than high order ARCH models much like how ARMA models are more efficient than high order AR models. In testing U.S. inflation data, Bollerslev found that a simple GARCH (1,1) model provided a marginally better fit and a more plausible learning mechanism than the eighth order ARCH model used in earlier literature. He derived that GARCH processes are heavy tailed, a property they share with ARCH processes, and established the formal conditions required for the process to be stationary. His findings showed that U.S. inflation was extremely volatile and hard to predict from the late 1940s to the mid-1950s, followed by a stable period in the 1960s, with a renewed increase in uncertainty starting with the 1974 oil crisis.

Airline stock returns during periods of geopolitical conflict are precisely the kind of series where ARCH effects are expected, where sharp, clustered spikes in volatility following conflict escalations or oil price jumps. We apply GARCH (1,1) models to each carrier's return series to estimate conditional volatility and test whether Iranian conflict driven oil shocks generate greater volatility in carriers with higher regional fuel exposure. The ARCH LM test,

rooted in Engle (1982), is used to confirm the presence of heteroskedasticity before estimation, and persistence estimates from Bollerslev (1986) are reported to characterize how long volatility shocks linger across the different regional carrier groups.

Caldara and Iacoviello (2022) develop a news-based measure of geopolitical risk and investigating how these risks influence macroeconomic activity and firm level decision making. The primary goal of the study is to create a consistent, real-time indicator of geopolitical tension, the Geopolitical Risk (GPR) index, and to use this index to answer how geopolitical factors shape economic outcomes. Specifically, the researchers examine how spikes in geopolitical risk aggregate variables like investment, employment, and stock prices are affected. Whether the threat of an adverse event has different economic consequences than the actual realization of that event. How is the relationship between geopolitical risk and the probability of economic disasters or downside risks to GDP growth? Also, how these risks affect individual firms and industries based on their specific levels of exposure.

The authors identify several key factors that motivate this research. First, stakeholder concerns like entrepreneurs, investors, and central bank officials consider geopolitical risk primary determinant of investment and market dynamics. A 2017 Gallup survey found that 75 percent of investors were worried about the economic impact of military and diplomatic conflicts. While many economic models suggest that geopolitical events can impact the macroeconomy through capital destruction, loss of life, or increased precautionary behavior, these factors had not previously been the subject of systematic empirical analysis. The study is motivated by the absence of a real time indicator that is consistent over time and captures geopolitical tensions perceived by the public, global investors, and policymakers. The researchers were motivated to distinguish between the anticipation threats, and the occurrence

acts of geopolitical events, as journalistic practices often intertwine these two components, making it difficult for researchers to isolate their individual effects.

The research utilized a news-based methodology to construct the GPR index, applying text search algorithms to millions of newspaper articles to quantify global and country specific tensions. The authors drew from several major datasets to build and validate their indices. With newspaper archives their core dataset consisted of approximately 25 million news articles published in leading English language newspapers since 1900. They then used 135,000 quarterly earnings calls of U.S. listed firms (2005-2019) to measure idiosyncratic geopolitical risk. For macroeconomic data, they utilized aggregate data on investment, employment, and stock prices, as well as historical GDP and disaster risk data for 26 countries.

The researchers employed a dictionary-based method to automate the measurement of geopolitical risk. The algorithm calculated the monthly share of articles discussing adverse geopolitical events. The search was organized into eight categories varying from low level threats to high level acts. The index was calculated by dividing the number of articles meeting the criteria by total number of articles in that month, then normalizing the result. They then split the headline index into two subcomponents. One being the Geopolitical Threats (GPT), focusing on risks and buildups. The other being Geopolitical Acts (GPA), focusing on the realization or escalation of events. They also created country specific indices by searching for the joint occurrence of geopolitical terms and country specific names.

To evaluate the economic impact of geopolitical risk, the authors used several econometric approaches. First, using VAR models to identify how shocks to the GPR index influenced aggregate U.S. variables like investment, employment, and stock prices. Then they employed quantile regressions to investigate how geopolitical risk affects the entire distribution

of future GDP growth, specifically looking at downside risks. Using disaster probability modeling, they use a linear probability specification to determine if spikes in the GPR index increased the likelihood of economic disasters. Then they used local projections to estimate how investment at individual firms responded to both aggregate and idiosyncratic geopolitical risks.

The research highlights that geopolitical risk is a significant driver of economic fluctuations, affecting both the overall macroeconomy and individual firm behavior through two primary channels, the anticipation of adverse events and their actual realization. Spikes in geopolitical risk led to significant and persistent declines in economic activity. A typical shock to the GPR index induces persistent declines in significant business fixed investment and employment. For example, a two standard deviation shock to the index can cause investment to bottom out at approximately 1.5 percent below the trend after about one year. These shocks are also followed by a decline in stock prices and a short-lived increase in financial uncertainty.

The study also found a strong link between geopolitical tensions and extreme negative economic outcomes. Higher levels of geopolitical risk are associated with a significantly higher probability of economic disasters. A one standard deviation increase in global GPR is associated with an 18 percent increase in the probability of a major economic disaster. Elevated risk is associated with larger downside risks to GDP growth. Specifically, a rise in the GPR index predicts a decline in the left tail of the GDP distribution that is four times larger than the average effect predicted by standard linear models. Also, the impact of geopolitical risk is not uniform across the economy as it depends heavily on a firm's specific exposure. Industries such as defense, petroleum, and transportation experience a significantly larger decline in investment than non-exposed industries.

The GPR index serves as our primary instrument for identifying Iranian conflict driven oil price shocks. Rather than relying on arbitrary event dummies for specific conflict dates. We look at monthly GPR values and Iranian conflict specific dates to isolate the precautionary demand component of oil price variation that is plausibly exogenous to airline specific stock price. Caldara and Iacoviello's finding that GPR shocks transmit to oil prices through precautionary demand directly motivates our identification strategy and provides the theoretical bridge between geopolitical conflict and fuel costs for the aviation sector.

Ullah et al. (2025) focuses on how oil price risk is transmitted to the airline industries of the U.S. and China during significant global crises. Specifically, they address the volatility of spillover relationship between oil returns and airline stocks during three distinct phases. First. The U.S.-China trade tensions, then the COVID-19 pandemic, and then the Russia-Ukraine war. They then go on to answering how does non-linearity and bad asymmetry affect the accuracy of volatility predictions in airline stocks. Also, what are the optimal hedging and diversification strategies using oil assets to mitigate risks for airline portfolios during market recessions?

The research was motivated by several critical economic factors and gaps in existing literature. The airline sector is extremely sensitive to disruptive global events that cause substantial market value losses. Because fuel is a primary operational expense, fluctuations in oil prices directly impact profitability. Historically, fuel expenses accounted for 20-24% of global airline costs (2015-2019), but most recently black swan events this share surged to 33% by the end of 2023, marking the most significant rise since World War II. Recent crises have drastically affected oil prices. For instance, a 41.2% single day drop during COVID-19 followed by an escalation to over \$100 per barrel during the Russian-Ukraine war. These shifts create information asymmetry and market instability. China and the U.S. are pivotal players in the

global oil market with different profiles. China is a massive importer of oil, approximately 73% of crude oil is imported. While the U.S. is a net exporter. Understanding how these superpowers navigate risk provides insights for global portfolio managers.

The research conducted by Ullah et al. (2025) utilizes a comprehensive dataset and several advance econometric models to analyze the transmission of oil price risk to airline stocks in the G2 economies. The study examines daily closing prices from January 2014 to November 2023. The data is then categorized into three distinct crisis periods such as the G2 trade tension, COVID-19, and Russian-Ukraine war. The study then focuses on the top five airlines from both China and the U.S. Daily WTI spot crude oil prices were retrieved using the Energy Information Administration (EIA).

The authors employed a methodological analysis to capture volatility spillovers and portfolio dynamics. Asymmetric GARCH was used to model time varying conditional correlation and capture bad asymmetry, where negative news impacts volatility more severely than positive news. The study then calculates several metrics to devise risk managements strategies. Including optimal weights to determine the ideal allocation of oil assets versus airline stocks in a \$1 portfolio to minimize risk without reducing expected returns. They also used hedge ratios to identify the cost effectiveness of short and long positions in oil to offset airline stock risk. Then they used hedging effectiveness to evaluate how much holding oil assets actually reduces portfolio risk compared to an unhedged portfolio.

The research provides a detailed analysis of how oil price volatility impacted the airline industry of China and the U.S. during three major global crises. Their main findings are categorized by the transmission of risk, portfolio management strategies, and the co-movement of these markets over time. The study found that both the oil market and the airline sectors in the

G2 economies are heavily influenced by their past volatility rather than just immediate news. During the COVID-19 pandemic, there was a heightened negative long run spillover from the oil market to Chinese airline stocks. In contrast, the Russian-Ukraine war harmed more significantly the U.S. aviation returns, with Delta and United Airlines experiencing negative spillovers. Also, the research confirmed the presence of bad asymmetry during these crises. This means that unknown negative news had a disproportionately larger and more damaging impact on airline stock prices than positive news of the same magnitude.

This paper is the most direct empirical predecessor to our study. Establishing oil price risk transmission to airline stocks is heterogeneous across national markets and crisis types. We extend Ullah et al. in two important directions. First, by expanding the regional coverage beyond the U.S. and China by including Europe, Asia, and Middle Eastern carriers. Secondly, by using specifically the Iran war oil shocks. Their findings of stronger Chinese sensitivity during war crises directly motivates our hypothesis that Asian carriers will exhibit larger responses to Hormuz disruption events than North American carriers.

Kilian et al. (2009) seek to provide a comprehensive analysis of the effects of different types of demand and supply shocks in the global crude oil market on the external balances of both crude oil-exporting and oil-importing economies. The primary goal was to quantify and differentiate the impact of various shocks. Specifically, oil supply disruptions, aggregate demand shocks, and oil market specific shocks.

The motivation for this research stems from several gaps in previous academic and policy discussions. Historically, most research on oil shocks focused on real economic growth and inflation in oil importing countries, while there was not much research on oil exporting countries. Recent spikes in oil prices and the emergence of large global external imbalances reignited

policy debates about the role of oil in determining these balances. The authors argue that traditional models incorrectly treat oil prices as purely exogenous and fail to recognize that the impact of an oil price shock depends heavily on its source. For example, a price increase driven by a booming global economy has different implications than one caused by supply disruptions. The researchers aim to determine if there is enough international risk sharing to cushion the domestic impact of shocks. They explore the financial channel of adjustment, which involves changes in asset prices and currency values, as a complement to the traditional trade channel.

The research uses an empirical approach focused on the period from 1975 to 2006 to analyze how different types of shocks in the global crude oil market affect external balances. The study includes 26 countries who are oil exporters and 12 countries who are major oil importers. The researchers examined several variables all normalized by nominal GDP: changes in net foreign assets, current trade balance, merchandise trade balance, oil trade balance, and capital gains.

The methodology involves a two-step structural approach designed to separate cause and effect by identifying the specific origin of oil price changes. The authors first estimated VAR models using monthly data to decompose fluctuations in the real price of oil into three distinct shocks. First, unexpected changes in global crude oil production. Then evaluating shifts in demand for all industrial commodities driven by the global business cycle. Lastly, they investigate changes in demand specific to oil, often reflecting precautionary demand due to concerns about future supply shortfalls. After evaluating the VAR models, the authors used an equation by equation distributed lag model to estimate the response of various external account measures to these annual shocks. This allowed them to trace the impact of a shock over a five-year horizon. Standard errors were constructed using a block bootstrap method to account for

serial correlation. Finally, the study employed a variance decomposition to determine the average importance of each shock in driving fluctuations in external balances over the entire sample period.

The research identifies several key findings regarding how different types of shocks in the crude oil market affect the external balance of oil exporting and oil importing nations. The most significant finding was that the source of oil price shock is critical. The impact on external accounts depends heavily on the underlying cause of the oil price change. Unexpected disruptions in oil production have small, short lived, and often statistically insignificant effects on the oil trade balance. However, oil market specific demand shocks cause large, persistent, and significant oil trade surpluses for exporters and deficits for importers. Also, aggregate demand shocks driven by the global business cycle result in what they call hump-shaped responses in the current account of oil exporters, peaking two to three years after the initial shock. The authors also document large systematic valuation effects through capital gains and losses on foreign assets that cushion the impact of oil demand shocks on net foreign asset positions, providing evidence of an intermediate degree of international financial integration.

Although Kilian et al. focus on external balances rather than equity markets, the paper is directly relevant to our study in two respects. First, we adopt the same three shock SVAR strategy as the backbone of our structural identification. Second, the finding that oil specific demand shocks driven by precautionary demand produce the largest and most persistent real economic effects directly motivates our focus on Iranian conflict driven shocks as the treatment of interest, since Hormuz disruption fears operate precisely through this precautionary demand channel. The external balance results also provide useful context for interpreting the cross

regional heterogeneity we expect to find in airline stock responses, as airline carriers in more oil import dependent economies face amplified downstream cost pressure from the same shocks.

Data

The empirical analysis combines daily airline equity prices, crude-oil prices, broad market returns, exchange rates, and geopolitical-event indicators. The airline sample contains nine publicly traded carriers divided equally across three regions. North America is represented by Delta Air Lines, United Airlines Holdings, and Southwest Airlines. Europe is represented by Ryanair Holdings, International Airlines Group, and Deutsche Lufthansa. Asia is represented by Air China, Singapore Airlines, and China Southern Airlines. Daily adjusted closing prices were obtained from Yahoo Finance. Adjusted prices were used because they account for stock splits, dividends, and other corporate actions, providing a consistent basis for calculating returns.

The energy-market variables are the daily spot prices of West Texas Intermediate, or WTI, and Brent crude oil obtained from the U.S. Energy Information Administration. WTI represents the primary U.S. crude-oil benchmark, while Brent serves as the broader international benchmark. The analysis also includes the daily S&P 500 return as a broad market control and daily returns on the EUR/USD, USD/JPY, and GBP/USD exchange rates. These control variables were obtained from Yahoo Finance. The S&P 500 controls for general equity-market movements that could affect airline stocks independently of oil prices, while the exchange-rate variables account for the international nature of the sample and the fact that airlines earn revenue, purchase fuel, and trade their shares in different currencies.

The requested sample begins on January 1, 2006. However, Delta Air Lines has the latest starting date among the nine carriers, with available price data beginning May 3, 2007. After restricting the data to dates shared by all nine airlines and merging the airline, oil, market, and exchange-rate series, the final modeling sample contains 4,404 observations from May 4, 2007, through June 8, 2026. The data are daily, although the observations represent common international trading dates rather than every calendar weekday. Restricting the sample in this manner prevents differences in national holidays and stock-exchange calendars from producing inconsistent cross-regional comparisons.

Airline, market, and exchange-rate returns are calculated 100 times the first difference of the natural logarithm of the adjusted closing price. These variables are therefore interpreted as approximate daily percentage returns. WTI and Brent are transformed using first differences in price levels, measured in U.S. dollars per barrel. First differences are used instead of logarithmic oil-price changes because WTI became negative in April 2020, making the logarithmic transformation undefined. Three equal-weighted regional return series are also constructed by averaging the daily returns of the three airlines within North America, Europe, and Asia.

The geopolitical-event dataset contains six Iran-related episodes: the May 2019 Fujairah tanker sabotage, the June 2019 Gulf of Oman tanker attacks, the September 2019 attacks on the Abqaiq and Khurais oil facilities, the January 2020 killing of Qasem Soleimani, Iran's April 2024 drone and missile attack on Israel, and Iran's October 2024 missile attack on Israel. The dates were documented using official records from the United Nations, U.S. Central Command, the U.S. Energy Information Administration, and the U.S. Department of Defense. Events that occurred on weekends or dates not shared by every stock market were assigned to the next available common trading date. The script creates both an event-day indicator and a seven-

observation event window extending from three common trading dates before through three common trading dates after each event.

Table 1: Summary Statistics

12_Descriptive_Statistics

Variable_ID	Variable	Observations	Mean	Median	Standard_Deviation	Minimum	Maximum	Skewness	Kurtosis	Excess_Kurtosis
DAL	Delta Air Lines	4404	0.0336	0.071	3.3431	-30.1003	23.5486	-0.2354	12.4802	9.4802
UAL	United Airlines Holdings	4404	0.0275	0.0179	4.1978	-45.8405	52.1986	-0.0995	21.4784	18.4784
LUV	Southwest Airlines	4404	0.0272	0.0638	2.3962	-20.3961	17.1412	-0.3747	9.2484	6.2484
RYA	Ryanair Holdings	4404	0.0355	0	2.4709	-25.5572	14.6378	-0.2631	9.9936	6.9936
IAG	International Airlines Group	4404	-0.0048	0.042	2.9797	-36.844	22.6997	-0.8891	15.9807	12.9807
LHA	Deutsche Lufthansa	4404	-0.0031	0	2.3993	-15.3021	18.129	0.0576	7.1544	4.1544
AIRCHINA	Air China	4404	-0.0015	0	3.0374	-37.5896	20.5852	-0.1797	11.8754	8.8754
SIA	Singapore Airlines	4404	0.0014	0	1.4499	-11.6123	13.0977	-0.0511	11.9216	8.9216
CHSOUTH	China Southern Airlines	4404	0.0135	0	3.2279	-19.0778	35.775	0.4324	9.6046	6.6046
North_America	North America (Regional Average)	4404	0.0294	0.067	3.0127	-24.8784	24.7303	-0.2171	12.3468	9.3468
Europe	Europe (Regional Average)	4404	0.0092	0.0296	2.2485	-17.3807	18.0136	-0.2081	8.5163	5.5163
Asia	Asia (Regional Average)	4404	0.0045	-0.0545	2.1655	-18.022	11.8076	-0.0868	7.7436	4.7436
D_WTI	Change in WTI	4404	0.0072	0.05	2.1596	-55.29	45.89	-1.8942	151.0834	148.0834
D_BRENT	Change in Brent	4404	0.0074	0.05	1.8356	-18	11.4	-0.6312	11.6504	8.6504
SP500_Return	S&P 500 Return	4404	0.0362	0.076	1.294	-12.7652	10.9572	-0.5168	14.266	11.266
EURUSD_Return	EUR/USD Return	4404	-0.0037	-0.0069	0.5856	-6.0011	5.8719	0.045	10.8357	7.8357
USDJPY_Return	USD/JPY Return	4404	0.0065	0.0114	0.6525	-6.057	6.3541	-0.1397	11.259	8.259
GBPUSD_Return	GBP/USD Return	4404	-0.0089	-0.0029	0.6069	-7.2289	5.2215	-0.6626	13.8892	10.8892

Table 1 reports the mean, standard deviation, minimum, maximum, skewness, and kurtosis of the financial variables. Average daily airline returns are small relative to their standard deviations, which is typical of financial return data. Ryanair Holdings has the highest average daily airline return at approximately 0.035 percent, followed closely by Delta Air Lines at approximately 0.034 percent. International Airlines Group, Deutsche Lufthansa, and Air China have slightly negative average daily returns over the sample. At the regional level, North America has the highest mean return at 0.029 percent per shared trading day, compared with 0.009 percent for Europe and 0.004 percent for Asia.

The results also demonstrate substantial differences in volatility. United Airlines Holdings has the highest airline-return standard deviation at approximately 4.20 percent, while

Singapore Airlines has the lowest at approximately 1.45 percent. The North American regional average is also more volatile than the European and Asian averages. These differences provide preliminary evidence that airline stocks do not respond uniformly to common economic, financial, and energy-market shocks.

The return distributions are strongly non-normal. Every airline series has kurtosis considerably greater than the normal distribution benchmark of 3. United Airlines has the highest airline kurtosis at approximately 21.48, while even the lowest airline value, approximately 7.15 for Deutsche Lufthansa, indicates substantial tail risk. WTI price changes are particularly extreme, with a minimum daily change of -55.29 dollars per barrel, a maximum of 45.89 dollars, skewness of -1.89 , and kurtosis of 151.08 . Brent changes are less extreme but remain negatively skewed and highly leptokurtic. These distributional characteristics indicate that unusually large observations occur more frequently than under a normal distribution. They also support the later use of Student-t innovations in the GARCH models, bootstrap confidence intervals in the VAR analysis, and heteroskedasticity- and autocorrelation-consistent standard errors in the local-projection models.

Figure 1: WTI and Brent Crude Oil Spot Prices

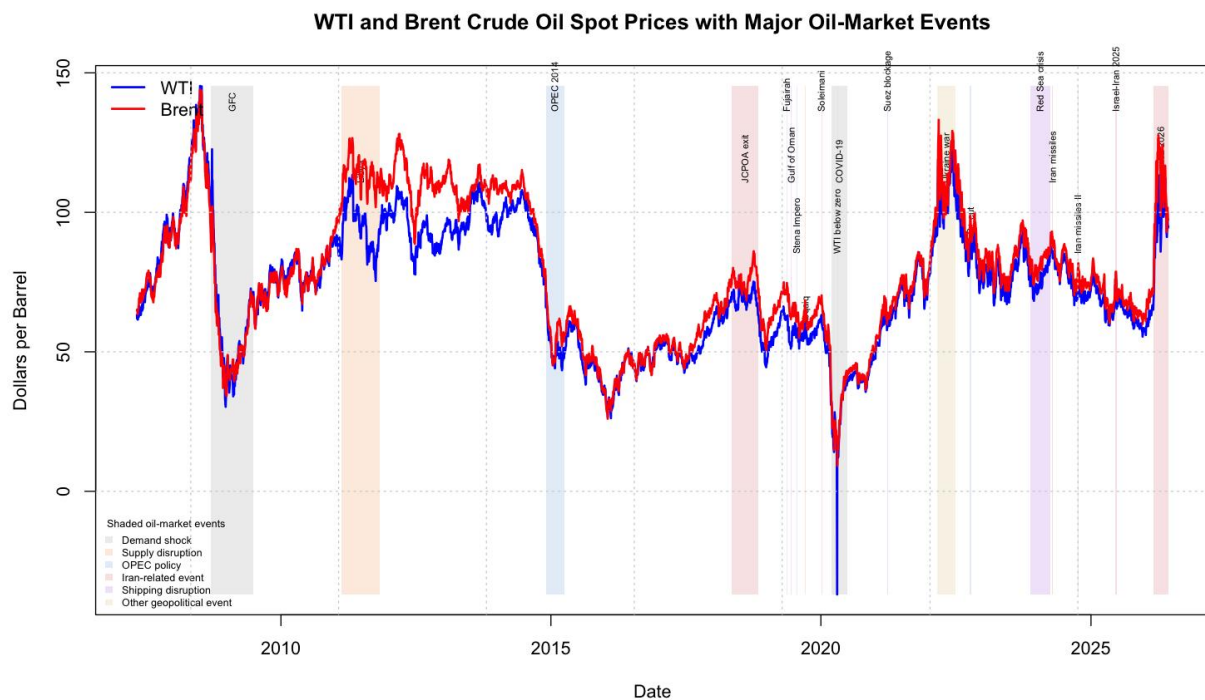
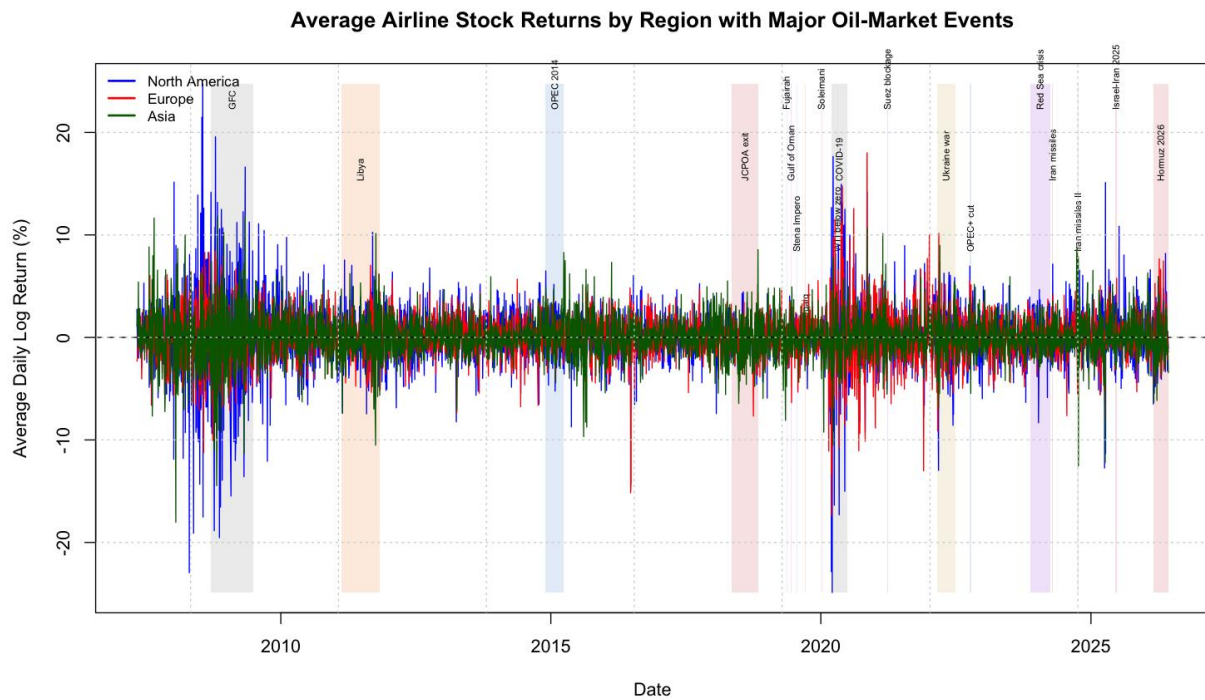


Figure 1 plots the WTI and Brent price levels. The two benchmarks generally move together, with Brent frequently trading above WTI, but several major disruptions are visible. Oil prices increased sharply before the 2008 global financial crisis and then declined rapidly during the recession. A second prolonged decline occurred between 2014 and 2016. The most pronounced structural break appears in April 2020, when the collapse in petroleum demand and severe storage constraints caused WTI to briefly fall below zero. Both benchmarks subsequently increased during the post-pandemic recovery and the 2021–2022 energy market disruption. A renewed increase is also visible near the end of the sample. Overall, the oil-price series exhibit substantial persistence in levels, major structural breaks, and time-varying volatility, but no stable seasonal pattern is visually apparent.

Figure 2: Average Airline Stock Returns by Region



The regional return series in Figure 2 fluctuates around zero and does not display an obvious deterministic trend or regular seasonal pattern. However, the series exhibits pronounced volatility clustering, with periods of relatively small movements followed by periods of unusually large positive and negative returns. The largest common movements occur during the 2008–2009 global financial crisis and the 2020 COVID-19 shock. Additional periods of elevated volatility are visible during later energy-market and geopolitical disruptions. North American airline returns display the largest extreme movements, which is consistent with their higher standard deviation in Table 1.

Figure 3: Regional Airline Return Indexes

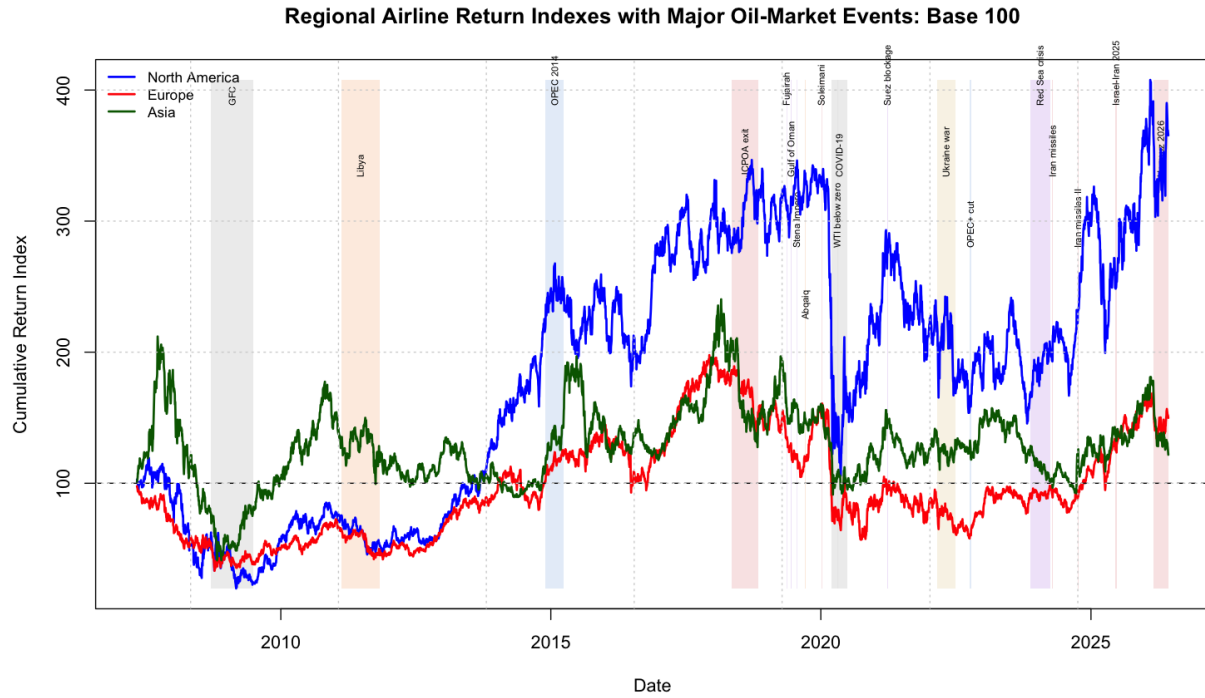


Figure 3 presents the cumulative regional return indexes; each normalized to 100 at the beginning of the sample. The North American index records the strongest long-run performance and ends the sample at approximately 365, compared with approximately 150 for Europe and 122 for Asia. Nevertheless, North America also experienced a severe decline during the COVID-19 shock. The European and Asian indexes recover more gradually and remain closer to their initial values for extended periods. The individual-airline figures for North America, Europe, and Asia further demonstrate that regional averages can conceal meaningful differences among carriers. For example, Delta, United, and Southwest follow distinct cumulative paths even though they operate within the same regional market.

Table 2: ADF and KPSS Stationarity Tests

14_Unit_Root_Test_Results

Variable_ID	Variable	ADF_Statistic	ADF_P_Value	PP_Statistic	PP_P_Value	KPSS_Statistic	KPSS_P_Value	ADF_Stationary	PP_Stationary	KPSS_Stationary
DAL	Delta Air Lines	-16.1158	0.01	-3963.9366	0.01	0.0868	0.1	Yes	Yes	Yes
UAL	United Airlines Holdings	-16.2069	0.01	-3921.7061	0.01	0.0572	0.1	Yes	Yes	Yes
LUV	Southwest Airlines	-15.969	0.01	-4768.4747	0.01	0.1003	0.1	Yes	Yes	Yes
RYA	Ryanair Holdings	-17.1182	0.01	-4147.5756	0.01	0.1008	0.1	Yes	Yes	Yes
IAG	International Airlines Group	-16.2104	0.01	-4374.3656	0.01	0.1216	0.1	Yes	Yes	Yes
LHA	Deutsche Lufthansa	-15.9603	0.01	-4162.3143	0.01	0.0552	0.1	Yes	Yes	Yes
AIRCHINA	Air China	-16.8378	0.01	-3971.4796	0.01	0.0362	0.1	Yes	Yes	Yes
SIA	Singapore Airlines	-16.2087	0.01	-4180.2322	0.01	0.1045	0.1	Yes	Yes	Yes
CHSOUTH	China Southern Airlines	-16.6068	0.01	-4156.8767	0.01	0.0627	0.1	Yes	Yes	Yes
North_America	North America (Regional Average)	-16.256	0.01	-4084.2795	0.01	0.0758	0.1	Yes	Yes	Yes
Europe	Europe (Regional Average)	-16.2555	0.01	-4160.7073	0.01	0.109	0.1	Yes	Yes	Yes
Asia	Asia (Regional Average)	-16.5184	0.01	-3953.2088	0.01	0.0326	0.1	Yes	Yes	Yes
D_WTI	Change in WTI	-15.1891	0.01	-4720.8889	0.01	0.0517	0.1	Yes	Yes	Yes
D_BRENT	Change in Brent	-14.4966	0.01	-4080.4512	0.01	0.0556	0.1	Yes	Yes	Yes
SP500_Return	S&P 500 Return	-16.603	0.01	-4744.6874	0.01	0.2618	0.1	Yes	Yes	Yes
EURUSD_Return	EUR/USD Return	-16.0628	0.01	-4400.1845	0.01	0.0454	0.1	Yes	Yes	Yes
USDJPY_Return	USD/JPY Return	-16.2105	0.01	-4226.5641	0.01	0.4146	0.0709	Yes	Yes	Yes
GBPUSD_Return	GBP/USD Return	-15.3053	0.01	-3976.6705	0.01	0.1217	0.1	Yes	Yes	Yes

Table 2 reports the Augmented Dickey–Fuller and KPSS tests for all continuous variables used in the empirical analysis. The null hypothesis of the ADF test is that the series contains a unit root, whereas the null hypothesis of the KPSS test is that the series is level stationary. The ADF test rejects the unit-root null for every airline return, regional return, oil-price change, market return, and exchange-rate return at the 1 percent significance level.

The KPSS statistics range from approximately 0.033 to 0.415, and all associated p-values exceed 0.05. Therefore, the KPSS tests fail to reject the null hypothesis of stationarity for every variable. Taken together, the ADF and KPSS results provide consistent evidence that the transformed series is stationary. Consequently, no additional differencing is required before estimating the VAR, SVAR, local-projection, GARCH, and panel models.

Empirical Methodology

This study employs four complementary empirical approaches to examine how geopolitical shocks to the oil market affect airline equity returns across three different regions. Event study analysis measures immediate market reactions to major oil market events, local

projections estimate the dynamic response of airline returns to oil price shocks, sign restricted structural vector autoregressions (SVARs) identify the transmission of structural oil shocks across the different regions. GARCH family models evaluate the effects of oil market disruptions on return volatility. Using multiple methodologies allows short run event specific effects and broader dynamic relationships to be analyzed while providing a robustness check across modeling frameworks.

Event Study

The event study framework evaluates abnormal airline stock performance surrounding major oil market events. Examples of this are OPEC policy, supply disruptions, sanctions, and military conflict. Abnormal returns are calculated relative to typical return behavior during a pre-event estimation period for each airline. Cumulative abnormal returns (CAR) are then computed over an event window extending three trading days after the event date.

$$CAR_i(t_1, t_2) = \sum_{t=t_1}^{t_2} AR_{it}$$

Where AR_{it} denotes the abnormal return for airline i on day t .

Event studies are particularly appropriate for this research question because financial markets rapidly incorporate new information into asset prices. Measuring cumulative abnormal returns around major geopolitical and oil-market events provides a direct assessment of how investors update expectations regarding airline profitability when energy-market conditions change.

Local Projections

To estimate the dynamic effects of oil price shocks this study employs the local projection methodology developed by Jordà (2005). Local projections estimate responses at each forecast horizon directly, which reduces sensitivity to model misspecification and allows greater flexibility.

For horizon h , the estimated specification is:

$$R_{t+h} = \alpha_h + \beta_h \Delta Oil_t + \gamma_h X_t + \varepsilon_t + h$$

In this specification, future airline returns are modeled as a function of contemporaneous oil-price shocks and a set of lagged control variables. Oil shocks are measured using daily changes in Brent and WTI crude oil prices, while the control variables account for broader market and macroeconomic conditions that may also influence airline equity performance. This approach is useful because it allows the effects of major geopolitical disruptions to be traced over time without imposing the restrictive assumptions associated with traditional VAR models.

Structural Vector Autoregression

To identify the transmission of structural oil shocks, a sign of restricted structural vector autoregression is estimated. The model includes daily changes in Brent and WTI crude oil prices with regional airline returns indices for North America, Europe, and Asia. The reduced form VAR is specified as:

$$Y_t = A_1 Y_{t-1} + A_2 Y_{t-2} + \dots + A_p Y_{t-p} + u_t$$

Where Y_t contains the endogenous variables are included in the system. A_1 through A_p are coefficient matrices. u_t represents reduced form innovations. Structural shocks are identified using sign restrictions. Oil-market shocks are required to generate positive contemporaneous

movements in oil prices, while broader market returns are restricted to be non-positive on impact. Airline returns remain unrestricted and are therefore determined by the data. This identification strategy allows economically meaningful oil shocks to be isolated without imposing the strong ordering assumptions required by a traditional Cholesky decomposition.

Impulse-response functions are computed over a twenty-trading-day horizon to evaluate both the magnitude and persistence of regional airline responses. Comparing responses across regions provides insight into whether airlines operating in different economic environments exhibit varying degrees of sensitivity to geopolitical oil-market disruptions.

Garch Volatility Models

Geopolitical oil market disruptions affect both expected returns and risk. Conditional volatility is modeled using standard GARCH(1,1) and asymmetric GJR-GARCH(1,1) specifications. The standard GARCH model is defined as:

$$\sigma^2_t = \omega + \alpha \varepsilon^2_{t-1} + \beta \sigma^2_{t-1}$$

Where σ^2_t denotes the conditional variance. The effect of recent shocks is captured by ε^2_{t-1} . The volatility persistence is captured by σ^2_{t-1} .

The GJR-GARCH specification extends the standard model by allowing negative shocks to have a different effect on the volatility compared to positive shocks of the same magnitude. This feature is critical as geopolitical crises, and sharp oil price increases may generate asymmetric market reactions. Model selections are conducted using the Akaike Information Criterion (AIC), with the preferred specification used to generate conditional volatility estimates.

The presence of volatility clustering and excess kurtosis in airline returns provides strong justification for employing GARCH family models.

Evaluation Framework

The primary objective of this paper is to evaluate the economic effects of geopolitical oil shocks on airlines. Model evaluation focuses on statistical significance, economic magnitude, and consistency across methodologies. For event studies, inference is based on cumulative abnormal returns and significance tests, while local projections and SVAR models are evaluated using impulse-response functions and confidence intervals to assess the persistence and regional heterogeneity of responses. Volatility models are assessed using AIC values and measures of volatility persistence. Employing multiple methodologies provides complementary evidence, and consistent findings across approaches strengthen conclusions regarding the impact of geopolitical oil-market disruptions on airline equity performance.

Results

The event study analysis reveals substantial heterogeneity in airline responses to major geopolitical and oil market events. Across the sample, the largest abnormal returns are associated with events that disrupted global supply chains or generated significant uncertainty regarding future oil supply. The Russian invasion of Ukraine generated one of the strongest reactions. European airlines experienced cumulative abnormal losses exceeding 17 percent over the event window. The results are consistent with the geographic proximity of European carriers to the conflict. Asian and North American carriers experienced losses as well but at a smaller magnitude. Significant negative reactions have also been observed during the periods of tension surrounding the Strait of Hormuz. One fifth of global petroleum consumption passes through this

corridor. This makes threats to shipping activity in the strait increase expectations of higher fuel costs and causes uncertainty.

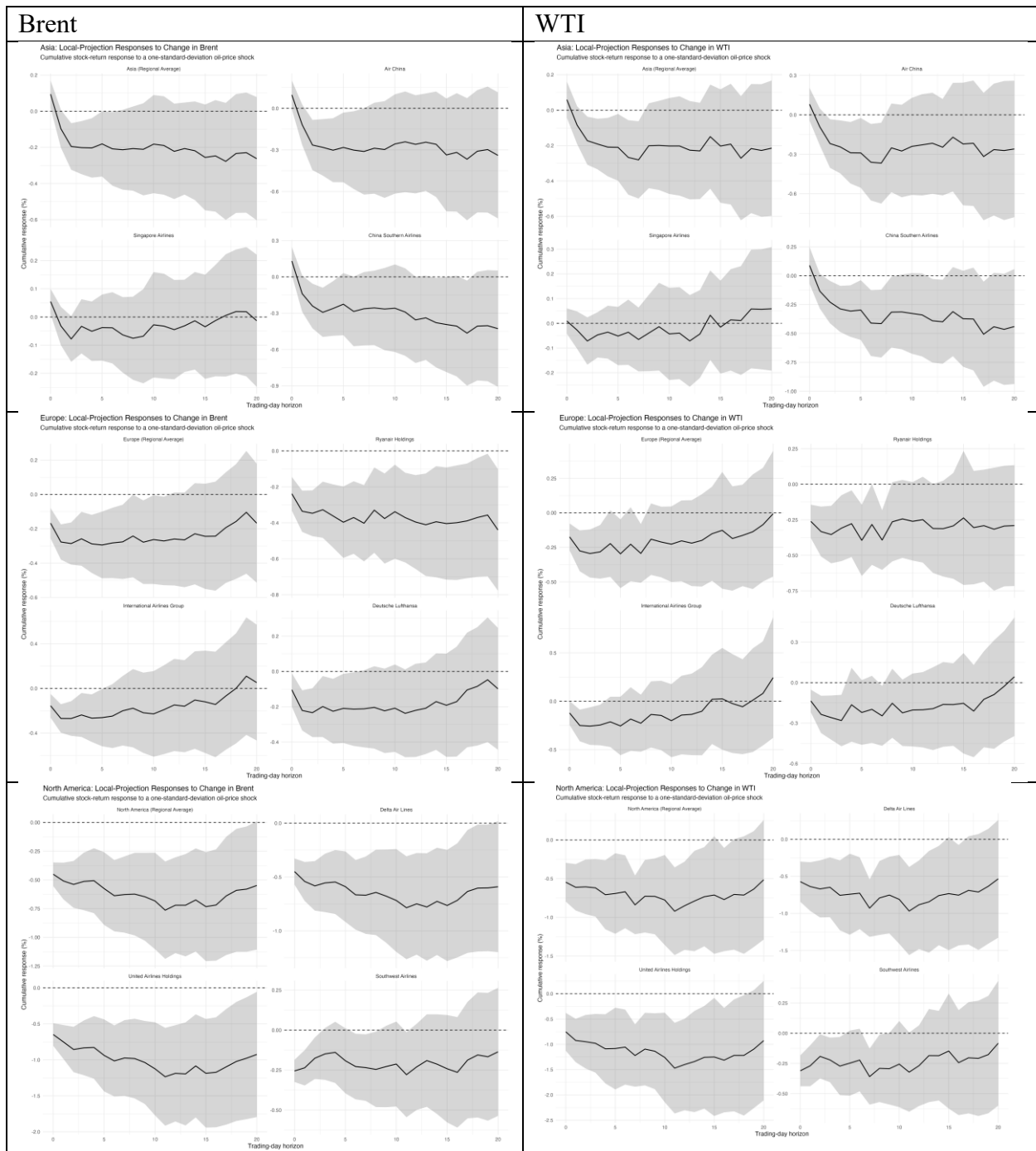
The event study framework also captures periods in which airlines benefited from favorable oil market developments. The 2014 OPEC decision to maintain high production levels saw airline stocks generate positive returns across multiple regions. European airlines experienced gains exceeding 13 percent while North American airlines gained more than 5 percent. The results are consistent with the significant role of fuel costs in airline profitability and provide evidence that the empirical framework captures economically meaningful responses to oil market developments.

The event study results demonstrate that airline equity markets respond strong to major geopolitical events with the magnitude varying across regions and by event type.

Dynamic Responses from Local Projections

Local projections provide additional insight into the persistence of airline responses following oil price shocks. The impulse response functions presented in Figure 4 are estimated using both Brent and WTI crude oil price changes.

Figure 4: Local Projection Impulse Responses of Regional Airline Returns to Oil Price Shock



Across all specifications, positive oil price shocks generate declines in airline stock returns. The estimated impulse responses indicate that the effects of oil shock are not limited to the day of the shock itself. Returns remain below baseline levels for several trading days

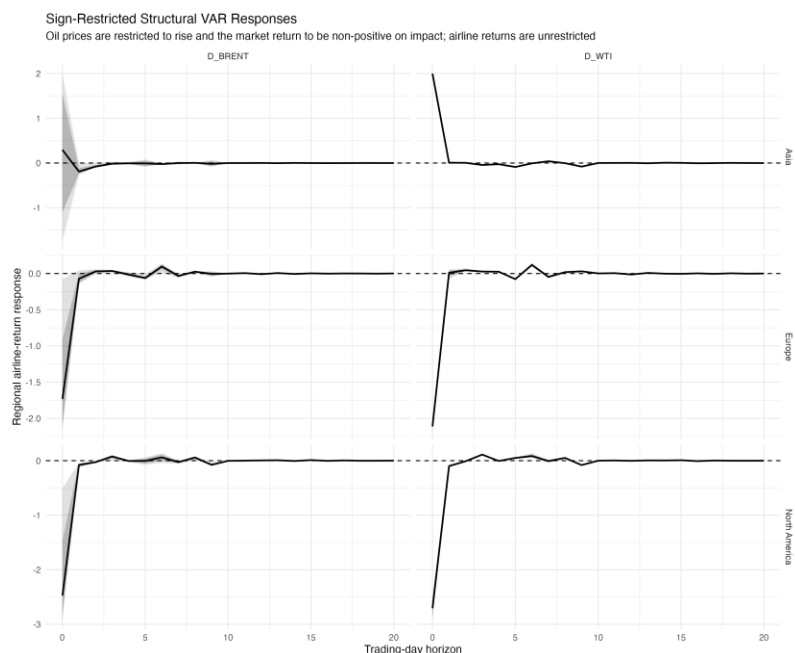
following the initial disturbance. Brent shock peak cumulative response reaches approximately -0.29 percent in Europe and -0.28 percent in Asia. Similar results are obtained using WTI shocks, suggesting that the findings are not driven by the choice of oil benchmark. An important feature of the local projection estimates is their persistence; they do not immediately revert to zero. The impulse responses remain negative throughout the twenty-day horizon. This pattern suggests that investors continue to revise expectations regarding future airline profitability after the initial shock occurs. The consistency of responses across both Brent and WTI specifications strengthens confidence in the robustness of the findings. The local projection results provide compelling evidence that oil price shocks negatively affect airline returns.

Structural VAR Results

To further isolate the effects of exogenous oil market disturbances, a sign restricted structural vector autoregression is estimated. Prior to structural identification, lag length was selected using standard information criteria and the stability of reduced form VAR was verified to ensure valid impulse response estimation.

Structural shocks are identified using sign restrictions that require oil price shocks to generate positive contemporaneous movements in crude oil prices while broader market returns remain non positive on impact. The resulting impulse response functions are presented in Figure 5.

Figure 5: Sign-Restricted SVAR Impulse Responses of Regional Airline Returns to Oil Price Shocks



The SVAR results indicate substantial regional heterogeneity in the transmission of oil shocks. North American airlines experience the largest contemporaneous declines, with estimated impact approaching -2.7 percent following identified WTI shocks. European airlines also experience significant negative responses, although the magnitude is somewhat smaller. Asian airlines exhibit considerably weaker reactions.

The impulse responses generally converge toward zero within several trading days. This indicates that financial markets incorporate information regarding oil shocks relatively quickly. The differences between regions are economically meaningful and broadly consistent with patterns observed in the event study analysis and the panel regressions.

The SVAR results suggest oil market disruptions impose the greatest burden on North American airlines. However, event-study results reveal that European carriers experience the largest losses during specific geopolitical crises, particularly the 2022 Russian invasion of Ukraine. Consequently, North American airlines appear more exposed to routine oil-price

fluctuations, while European airlines exhibit greater vulnerability to certain regional geopolitical disruptions.

Panel Regression Results

Panel regression estimates provide the strongest statistical evidence regarding regional differences in airline sensitivity to oil price shocks. Using North America as the baseline category, both WTI and Brent oil shocks are associated with significant declines in airline stock returns. For WTI, the estimated oil shock coefficient is -0.1728 ($p < 0.001$), while the Brent specification produces a coefficient of -0.1681 ($p < 0.001$). These results indicate that increases in oil prices have an economically meaningful effect on airline equity performance.

Regional interaction terms reveal substantial heterogeneity in these responses. The positive and statistically significant interaction coefficients for Asia and Europe indicate both regions to be less sensitive to oil price shocks than North America. The Asia coefficient equals 0.158 in the WTI specification and 0.1812 in the Brent specification. This offsets much of the negative baseline effect. European interaction coefficients are positive and significant in both models similarly. The formal tests confirm these differences.

The control variables behave as expected. S&P 500 returns are strongly and positively associated with airline returns, with coefficients ranging from approximately 0.76 to 0.77 ($p < 0.001$). In addition, the event-window indicator is negative and statistically significant in the clustered specification, suggesting that major geopolitical events generate abnormal losses even after controlling for oil-price movements and broader market conditions.

Volatility Dynamics

While the preceding analyses focus on the effects of oil shocks on expected returns, GARCH-family models are used to evaluate their effects on financial market risk. Prior diagnostic testing indicated significant volatility clustering and excess kurtosis in airline returns. This provides strong justification for the use of conditional volatility models.

Model selection was performed using the AIC, and the preferred specification for each airline was subsequently used to generate conditional volatility estimates. Most airlines were best modeled using asymmetric GJR-GARCH specifications, indicating that negative shocks have a larger effect on future volatility than positive shocks of similar magnitude. Only Southwest Airlines and Singapore Airlines were better described by the symmetric GARCH specification.

The volatility estimates reveal clear evidence of clustering, with periods of elevated volatility concentrated around major economic and geopolitical disruptions. The largest volatility spikes occur during the Global Financial Crisis, the COVID-19 pandemic, the Russian invasion of Ukraine, and recent Middle Eastern shipping disruptions. These episodes coincide with periods of heightened uncertainty regarding both economic activity and future oil prices.

North American airlines exhibit the highest average conditional volatility (2.54), followed by Asian (2.43) and European (2.32) carriers. However, conflict-period volatility increases are most pronounced in Europe, where the conflict-to-normal volatility ratio reaches 1.06. Consistent with the event-study results, European airlines appear particularly vulnerable during periods of geopolitical disruption, especially conflicts with direct implications for regional energy markets.

Additional statistical tests reveal that Lufthansa is the only airline exhibiting a statistically significant difference between conflict-period and non-conflict volatility levels, while Ryanair displays marginal significance. Although volatility clustering is present across all

airlines, these results suggest that conflict-related volatility effects are concentrated among a subset of European carriers rather than uniformly distributed across the industry.

Summary of Findings

Across all empirical approaches there is a consistent pattern which emerges. Geopolitical oil market disruptions generate negative effects on airline equity performance although the magnitude of those effects varies by region. Event study estimates show that conflicts and supply distributions can produce substantial abnormal losses, while favorable oil market developments generate significant gains. Local projects and SVAR models demonstrate that positive oil shocks reduce airline returns, although statistically significant regional differences are more clearly identified in the panel regression framework than in the local projection estimates.

The panel regressions provide the strongest evidence regarding regional heterogeneity. GARCH models indicate that geopolitical shocks increase financial market uncertainty through persistent and asymmetric volatility dynamics. The consistency of the findings across event study analysis, panel regressions, SVAR models, and GARCH estimation provides convincing evidence that oil market disruptions represent an important source of both return and volatility risk for the airline industry.

Conclusion

This paper examined how geopolitical oil market disruptions affect airline equity performance across North America, Europe, and Asia. Using a sample of major publicly traded airlines from 2007 to 2026, the analysis combined event study methods, local projections, sign restricted structural vector autoregressions, panel regressions and GARCH family volatility

models to evaluate both the return and volatility effects of oil price shocks. The objective was to determine whether airline responses differ across regions.

The results consistently indicate that oil market disruptions negatively affect airline stock returns. Event study estimates show that major geopolitical events, such as the Russian invasion of Ukraine and the Iran conflict have generated substantial abnormal losses for airline stocks. Conversely, favorable oil-market developments such as the 2014 OPEC supply expansion produced significant positive abnormal returns, supporting the importance of fuel costs in determining airline profitability.

The strongest statistical evidence regarding regional differences comes from the panel regression analysis. Both Brent and WTI oil shocks are associated with significant declines in airline returns, while regional interaction terms indicate that North American airlines are generally more sensitive to oil-price shocks than their European and Asian counterparts. Sign-restricted SVAR results support this conclusion, showing the largest contemporaneous responses among North American carriers. However, the event-study results demonstrate that European airlines experienced the largest losses during specific geopolitical crises, particularly the 2022 Russian invasion of Ukraine. Taken together, these findings suggest that North American airlines are more exposed to routine oil-price fluctuations, while European airlines appear particularly vulnerable to certain regional geopolitical disruptions.

The volatility analysis further demonstrates that geopolitical events influence not only expected returns but also financial market risk. GARCH-family models reveal substantial volatility persistence across all airlines, with most carriers exhibiting asymmetric responses in which negative shocks generate disproportionately large increases in future volatility. While North American airlines display the highest average conditional volatility, conflict-related

volatility increases are most pronounced among European carriers, further reinforcing the importance of regional differences in exposure to geopolitical risk.

This study contributes to the existing literature by extending research on oil-price shocks and airline performance to a comparative regional framework. Whereas much of the previous literature focuses on individual countries, specific events, or aggregate airline indices, this paper directly compares airline responses across North America, Europe, and Asia using multiple empirical methodologies. The consistency of findings across event studies, panel regressions, SVAR models, local projections, and volatility models strengthens confidence in the conclusion that oil-market disruptions remain an important determinant of airline equity performance.

The results also have implications for investors and risk managers. Airlines remain highly exposed to energy-market developments despite improvements in hedging practices and operational efficiency. Investors should therefore consider regional differences when evaluating airline-sector risk. North American carriers appear more sensitive to broad oil-price shocks, while European airlines may face elevated downside risk during geopolitical conflicts that directly affect regional energy markets.

Several limitations should be acknowledged. The analysis focuses on publicly traded airlines and therefore may not fully capture the behavior of privately held carriers or state-owned airlines. Geopolitical events often occur simultaneously with broader economic and financial disturbances, making it difficult to completely isolate the effects of oil-market shocks. Although sign-restricted SVARs and event-study methods help address this issue, some degree of identification of uncertainty remains. Differences in fuel-hedging practices, route structures, and firm-specific operating characteristics are not directly observed and may contribute to the regional patterns identified in the analysis.

Future research could investigate firm-level hedging strategies, route-network exposure, and fuel-cost pass-through mechanisms to better understand why airline responses differ across regions. Additional work could also examine the interaction between geopolitical risk and other industry-specific factors such as environmental regulations, sustainable aviation fuel adoption, and changing patterns of international air travel. As geopolitical tensions and energy-market uncertainty continue to shape the global economy, understanding the relationship between oil shocks and airline performance will remain a key area of research for both academics and practitioners.

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